Ahmadh Hassan

ahmadhhassan05@gmail.com| 848.482.9198 | linkedin.com/in/ahmadhhassan/

EDUCATION

Rutgers University-New Brunswick, NJ
B.S. Electrical & Computer Engineering; B.S. Applied Physics; B.S. Mathematics May 2027
GPA: 3.8/4.0
Honors: IEEE Eta Kappa Nu Electrical Honor Society, Sigma Pi Sigma Physics Honor Society, James J. Slade Scholar, NASA Space Grant recipient, Dean's List 6x
Related Coursework:
UG ECE: Computer Architecture, Electronic Devices, Digital Signal Processing, Linear Systems and Signals
UG Math & Physics: Quantum Mechanics, Statistical Mechanics, Partial Differential Equations, Abstract Linear Algebra, Probability Theory
Graduate: Solid State Electronics, Microelectronic Processing, Special Study in Advanced Math, Quantum Mechanics, Semiconductor Devices

EXPERIENCE

Argonne National Laboratory June 2026-August 2026
Quantum Computing Research Intern | U.S. Department of Energy SULI | Mentor: Dr. Yizhong Huang

- Design superconducting microwave resonators for electron-on-solid-neon quantum-device experiments and develop lithography-ready device layouts
- Use Ansys HFSS eigenmode and driven-modal simulations to evaluate resonant frequencies, S-parameters, electric- and magnetic-field distributions, quality factors, and mesh and frequency convergence.
- Compare simulated resonator behavior with analytical estimates and published designs, then iterate device geometry to improve mode localization and expected performance.
- Characterize erbium-doped CaWO4 and Y2O3 powder samples using scanning electron microscopy and energy-dispersive X-ray spectroscopy to evaluate particle morphology and elemental composition for quantum-material studies.

Condensed Matter Experiment (CMX) Research May 2025-Present
Undergraduate Research Assistant| Advisor: Prof. Jak Chakhalian

- Lead an independent honors thesis on the design and fabrication of gated Eu2Ir2O7 pyrochlore-iridate Hall-bar devices for cryogenic magnetotransport measurements down to 2 K.
- Investigate nonlinear current-voltage characteristics and second-harmonic transport response under applied magnetic fields.
- Fabricate oxide-heterostructure and topological-insulator devices using optical lithography, electron-beam lithography, plasma etching, sample preparation, and wire bonding.
- Perform resistivity, Hall-effect, and magnetoresistance measurements on Bi2Se3 and Bi2Te3 devices using Quantum Design PPMS systems and lock-in amplifiers.
- Conduct torque-magnetometry and vibrating-sample-magnetometry measurements on Cr2Te3 to examine field-dependent magnetization and magnetic anisotropy.
- Analyze transport and magnetic data in Origin and optimize measurement and filtering parameters, improving signal-to-noise ratio by approximately 100-fold.

Device-independent Quantum Key Distribution Research (DI-QKD) June 2024 - August 2024
Undergraduate Research Assistant | Advisor: Prof. Emina Soljanin

- Studied device-independent quantum key distribution under untrusted-device and eavesdropping scenarios.
- Implemented the CHSH game in Qiskit and Python to model entanglement correlations and protocol behavior under noise.
- Developed simulations to evaluate protocol robustness under adversarial conditions.

OPENSOURCE SOFTWARE

QResAudit — Open-Source Electromagnetic Simulation Validation Framework
Python, PyAEDT, HFSS, HDF5, Pydantic, scikit-rf, GitHub Actions

- Developed an open-source framework for reproducible electromagnetic simulation workflows, extracting HFSS field data, network parameters, convergence metrics, and provenance metadata into portable evidence archives for quantum-device simulations.
- Designed a solver-independent validation pipeline enabling transparent verification of superconducting resonator and computational electromagnetics workflows independent of proprietary simulation environments.

POSTERS & PRESENTATIONS

Hassan, A. "Design and Simulation of a Superconducting Resonator for Single Electron Spin Detection " Poster presentation at Argonne National Laboratory, Lemont, IL, August 2026

Hassan, A. "Electrical Interfacing for a 6U CubeSat." Poster presented at the New Jersey Space Grant Consortium Summer Internship Poster Session, Rutgers University, New Brunswick, NJ, August 2024

Hassan, A. "A Brief Introduction to Differential Geometry." Student presentation, Rutgers Mathematics Directed Reading Program, Spring 2025

Hassan, A. "A Brief Overview of Point-Set Topology." Student presentation, Rutgers Mathematics Directed Reading Program, Fall 2024

ENGINEERING EXPERIENCE

NASA via NJSGC Electronics & Power Intern June 2024 - August 2024
Research Title: Electrical Interfacing for a 6U CubeSat New Brunswick, NJ

- Designed and optimized electrical systems for a 6U CubeSat, including power budget redesign (+66.7% power generation) and integration of flight computer, power system, and thermal control system
- Collaborated on PCB design for CubeSat components, ensuring compliance with system-level validation standards
- Managed construction and implementation of Thermal Control System, optimized heat dissipation and thermal management

TECHNICAL & ACADEMIC PROJECTS

Renormalization Groups in Graphene January 2026-Present

- Study renormalization-group methods in interacting graphene systems and quantum statistical mechanics through guided reading and written exposition.

Presentation Project – Evolution of Vorticity in the Euler and Navier-Stokes Equations January 2025 - May 2025

- Semester-long project exploring the evolution of vorticity in 2D and 3D incompressible fluid models, including Euler and Navier-Stokes equations
- Derived and analyzed key PDEs governing vorticity, and discussed the Biot–Savart law, stream functions, and vortex stretching in higher dimensions.
- Connected results to Burgers' equation to gain simplified insight into nonlinear transport and stretching phenomena

LEADERSHIP EXPERIENCE

Space Technology Association at Rutgers September 2023 - Present
Power Team Lead New Brunswick, NJ

- Managing a team of 10 to optimize electronics efficiency for a satellite
- Investigating inhibit systems by prototyping and simulating in VHDL and Verilog
- Incorporated attitude parameters into an STK model, increasing power generation from 12W to 20W
- Conducted Power and RF communication simulations and investigated EM radiation on component health

Rutgers University Quantum Computing Group May 2024 - Present
President/Co-Founder New Brunswick, NJ

- Founded and led a student organization focused on quantum computing, growing membership to 70+ students
- Organized workshops and lectures to educate members and the public on quantum computing fundamentals
- Collaborated with professors, researchers, and companies to promote quantum computing research and outreach

THESIS/CAPSTONE

Senior Honors Thesis & J.J. Slade Research Fellowship August 2025-Present
Advisor: Prof. Chakhalian Rutgers Department of Physics and Astronomy

- Independently designing and fabricating perovskite heterostructures for cryogenic transport studies
- Investigating second harmonic generation and non-linear I–V characteristics under applied magnetic fields
- Using SPUV Lithography and Plasma etching to create hall bar device on Europium Iridate

Senior Capstone August 2026-Present
Advisor: Prof. Shriram Ramanathan Rutgers Department of Electrical Engineering

- Developing inorganic neuron and synapse devices using quantum materials with programmable switching dynamics and neuromorphic functionality
- Investigating device architectures for scalable artificial neural network hardware

- **Python**
- **Ansys HFSS**
- **Lithography**
- **MATLAB**
- **PyTorch**
- **Plasma Etching**
- **Verilog**
- **Altium Designer**
- **Cryogenic Measurements**
- **PyAEDT**
- **AGI STK**
- **PPMS Characterization**